LARGEST OPEN WHERE H HAS P DESCENDS ALONG THE PROJECTION WITH SECTION, FOR A GROUPOID WHOSE SOURCE AND TARGET HAVE R AND A BASE CHANGE G HAVING Q

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-groupoids.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- The pullback diagram of the groupoid recording that c composed with (i, 1) over s equals h over t with both squares cartesian (more-groupoids chapter, equation equation-pull, canonical tag 04LG; kind equation). cited at proof lines 525-526. Its properties, combined with the first cartesian diagram of the proof, give the second commutative cartesian diagram identifying pr_01^{-1}(W(P, h)) and pr_02^{-1}(W(P, h)) as the largest open where c composed with (i, 1) has P.

Source reference. Original equation 04LG: sources/stacks-chow/tags.txt, source lines 6007–6007. Cited at proof lines 525-526: the properties of diagram (equation-pull), combined with the first cartesian diagram, produce the second commutative cartesian diagram relating pr_01, pr_02 to c composed with (i, 1) and pr_1, h to s, on which the invariance of W(P, h) rests.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c, e, i) be a groupoid over S. Let $g : U' \to U$ be a morphism of schemes. Denote h the composition*

$$h : U' \times_{g,U,t} R \xrightarrow[pr_1]{} R \xrightarrow[s]{} U.$$

Let $\mathcal{P}, \mathcal{Q}, \mathcal{R}$ be properties of morphisms of schemes. Assume

(1) *$\mathcal{R} \Rightarrow \mathcal{Q}$,*
(2) *$\mathcal{Q}$ is preserved under base change and composition,*
(3) *for any morphism $f : X \to Y$ which has $\mathcal{Q}$ there exists a largest open $W(\mathcal{P}, f) \subset X$ such that $f|_{W(\mathcal{P},f)}$ has $\mathcal{P}$, and*
(4) *for any morphism $f : X \to Y$ which has $\mathcal{Q}$, and any morphism $Y' \to Y$ which has $\mathcal{R}$ we have $Y' \times_Y W(\mathcal{P}, f) = W(\mathcal{P}, f')$, where $f' : X_{Y'} \to Y'$ is the base change of f.*

If s, t have $\mathcal{R}$ and g has $\mathcal{Q}$, then there exists an open subscheme $W \subset U'$ such that $W \times_{g,U,t} R = W(\mathcal{P}, h)$.

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Proof. Note that the following diagram is commutative

$$\begin{array}{ccc} U' \times_{g,U,t} R \times_{t,U,t} R & \xrightarrow[\mathrm{pr}_{12}]{} & R \times_{t,U,t} R \\ {\scriptstyle \mathrm{pr}_{01}}\downarrow \downarrow{\scriptstyle \mathrm{pr}_{02}} & & {\scriptstyle \mathrm{pr}_0}\downarrow \downarrow{\scriptstyle \mathrm{pr}_1} \\ U' \times_{g,U,t} R & \xrightarrow{\mathrm{pr}_1} & R \end{array}$$

with both squares cartesian (this uses that the two maps $t \circ \mathrm{pr}_i : R \times_{t,U,t} R \to U$ are equal). Combining this with the properties of diagram (04LG) we get a commutative diagram

$$\begin{array}{ccc} U' \times_{g,U,t} R \times_{t,U,t} R & \xrightarrow[c \circ (i,1)]{} & R \\ {\scriptstyle \mathrm{pr}_{01}}\downarrow \downarrow{\scriptstyle \mathrm{pr}_{02}} & & {\scriptstyle t}\downarrow \downarrow{\scriptstyle s} \\ U' \times_{g,U,t} R & \xrightarrow{h} & U \end{array}$$

where both squares are cartesian.

Assume s, t have $\mathcal{R}$ and g has $\mathcal{Q}$. Then h has $\mathcal{Q}$ as a composition of s (which has $\mathcal{R}$ hence $\mathcal{Q}$) and a base change of g (which has $\mathcal{Q}$). Thus $W(\mathcal{P}, h) \subset U' \times_{g,U,t} R$ exists. By our assumptions we have $\mathrm{pr}_{01}^{-1}(W(\mathcal{P}, h)) = \mathrm{pr}_{02}^{-1}(W(\mathcal{P}, h))$ since both are the largest open on which $c \circ (i, 1)$ has $\mathcal{P}$. Note that the projection $U' \times_{g,U,t} R \to U'$ has a section, namely $\sigma : U' \to U' \times_{g,U,t} R$, $u' \mapsto (u', e(g(u')))$. Also via the isomorphism

$$(U' \times_{g,U,t} R) \times_{U'} (U' \times_{g,U,t} R) = U' \times_{g,U,t} R \times_{t,U,t} R$$

the two projections of the left hand side to $U' \times_{g,U,t} R$ agree with the morphisms pr_{01} and pr_{02} on the right hand side. Since $\mathrm{pr}_{01}^{-1}(W(\mathcal{P}, h)) = \mathrm{pr}_{02}^{-1}(W(\mathcal{P}, h))$ we conclude that $W(\mathcal{P}, h)$ is the inverse image of a subset of U, which is necessarily the open set $W = \sigma^{-1}(W(\mathcal{P}, h))$. □